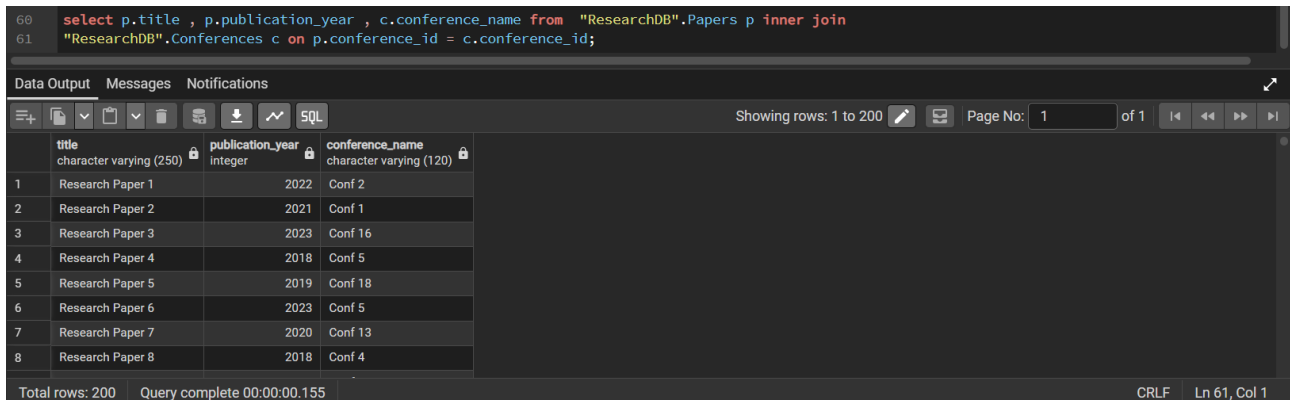


Student id = 202612002

Group - A

Q1. Display the paper title and publication year along with the conference name.

```
select p.title , p.publication_year , c.conference_name from "ResearchDB".Papers p inner join  
"ResearchDB".Conferences c on p.conference_id = c.conference_id;
```



The screenshot shows a SQL query execution interface. The query is: `select p.title , p.publication_year , c.conference_name from "ResearchDB".Papers p inner join "ResearchDB".Conferences c on p.conference_id = c.conference_id;`. The results are displayed in a table with 3 columns: `title` (character varying (250)), `publication_year` (integer), and `conference_name` (character varying (120)). The table contains 8 rows of data. The interface also shows a status bar at the bottom indicating 'Total rows: 200' and 'Query complete 00:00:00.155'.

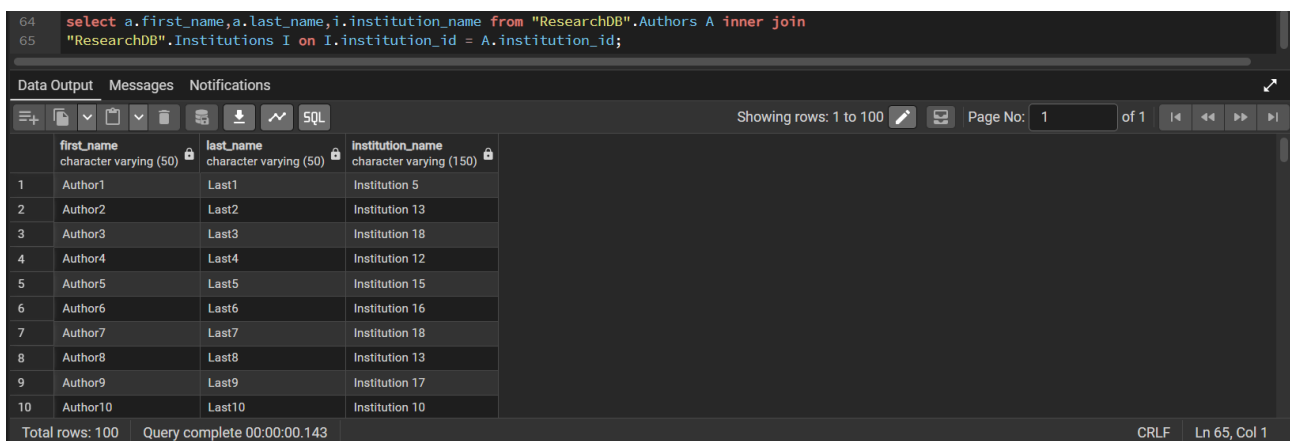
	title character varying (250)	publication_year integer	conference_name character varying (120)
1	Research Paper 1	2022	Conf 2
2	Research Paper 2	2021	Conf 1
3	Research Paper 3	2023	Conf 16
4	Research Paper 4	2018	Conf 5
5	Research Paper 5	2019	Conf 18
6	Research Paper 6	2023	Conf 5
7	Research Paper 7	2020	Conf 13
8	Research Paper 8	2018	Conf 4

Total rows: 200 Query complete 00:00:00.155 CRLF Ln 61, Col 1

Tuple Count = 200

Q2. Display the author name and institution name.

```
select a.first_name,a.last_name,i.institution_name from "ResearchDB".Authors A inner join  
"ResearchDB".Institutions I on I.institution_id = A.institution_id;
```



The screenshot shows a SQL query execution interface. The query is: `select a.first_name,a.last_name,i.institution_name from "ResearchDB".Authors A inner join "ResearchDB".Institutions I on I.institution_id = A.institution_id;`. The results are displayed in a table with 3 columns: `first_name` (character varying (50)), `last_name` (character varying (50)), and `institution_name` (character varying (150)). The table contains 10 rows of data. The interface also shows a status bar at the bottom indicating 'Total rows: 100' and 'Query complete 00:00:00.143'.

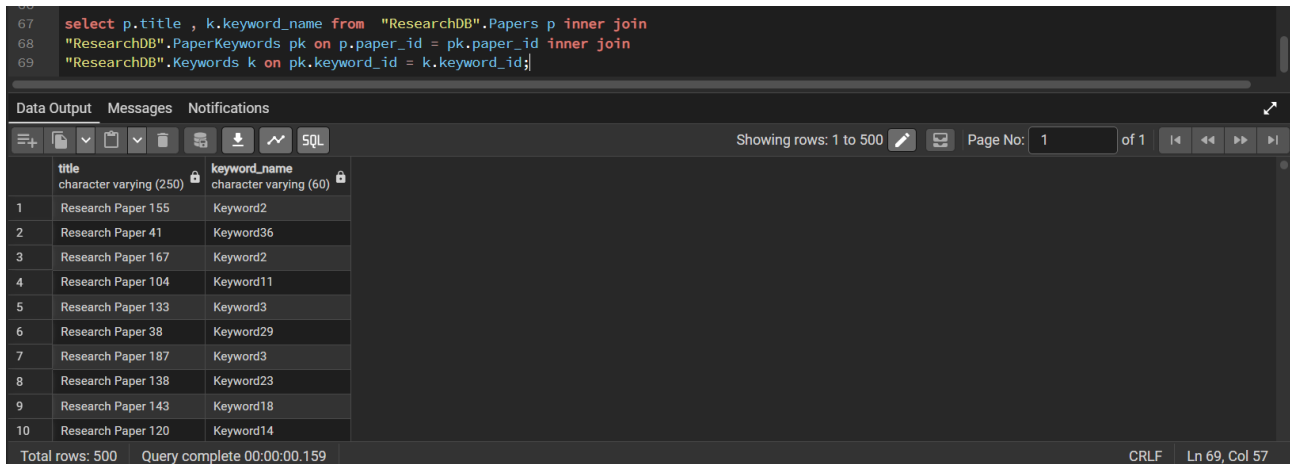
	first_name character varying (50)	last_name character varying (50)	institution_name character varying (150)
1	Author1	Last1	Institution 5
2	Author2	Last2	Institution 13
3	Author3	Last3	Institution 18
4	Author4	Last4	Institution 12
5	Author5	Last5	Institution 15
6	Author6	Last6	Institution 16
7	Author7	Last7	Institution 18
8	Author8	Last8	Institution 13
9	Author9	Last9	Institution 17
10	Author10	Last10	Institution 10

Total rows: 100 Query complete 00:00:00.143 CRLF Ln 65, Col 1

Tuple Count = 100

Q3. Display the paper title and keyword.

```
select p.title , k.keyword_name from "ResearchDB".Papers p inner
join "ResearchDB".PaperKeywords pk on p.paper_id = pk.paper_id inner
join "ResearchDB".Keywords k on pk.keyword_id = k.keyword_id;
```



```
67 select p.title , k.keyword_name from "ResearchDB".Papers p inner join
68 "ResearchDB".PaperKeywords pk on p.paper_id = pk.paper_id inner join
69 "ResearchDB".Keywords k on pk.keyword_id = k.keyword_id;
```

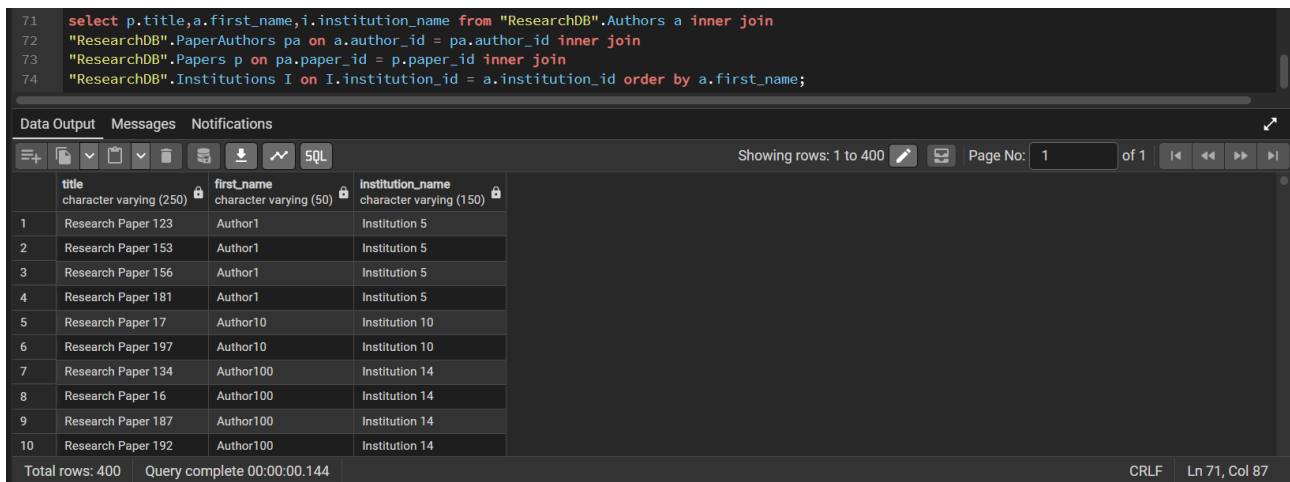
	title character varying (250)	keyword_name character varying (60)
1	Research Paper 155	Keyword2
2	Research Paper 41	Keyword36
3	Research Paper 167	Keyword2
4	Research Paper 104	Keyword11
5	Research Paper 133	Keyword3
6	Research Paper 38	Keyword29
7	Research Paper 187	Keyword3
8	Research Paper 138	Keyword23
9	Research Paper 143	Keyword18
10	Research Paper 120	Keyword14

Total rows: 500 Query complete 00:00:00.159 CRLF Ln 69, Col 57

Tuple count = 500

Q4. Display the paper title, author name, and institution name.

```
select p.title,a.first_name,i.institution_name from "ResearchDB".Authors a inner
join "ResearchDB".PaperAuthors pa on a.author_id = pa.author_id inner join "ResearchDB".Papers
p on pa.paper_id = p.paper_id inner join "ResearchDB".Institutions I on I.institution_id =
a.institution_id order by a.first_name;
```



```
71 select p.title,a.first_name,i.institution_name from "ResearchDB".Authors a inner join
72 "ResearchDB".PaperAuthors pa on a.author_id = pa.author_id inner join
73 "ResearchDB".Papers p on pa.paper_id = p.paper_id inner join
74 "ResearchDB".Institutions I on I.institution_id = a.institution_id order by a.first_name;
```

	title character varying (250)	first_name character varying (50)	institution_name character varying (150)
1	Research Paper 123	Author1	Institution 5
2	Research Paper 153	Author1	Institution 5
3	Research Paper 156	Author1	Institution 5
4	Research Paper 181	Author1	Institution 5
5	Research Paper 17	Author10	Institution 10
6	Research Paper 197	Author10	Institution 10
7	Research Paper 134	Author100	Institution 14
8	Research Paper 16	Author100	Institution 14
9	Research Paper 187	Author100	Institution 14
10	Research Paper 192	Author100	Institution 14

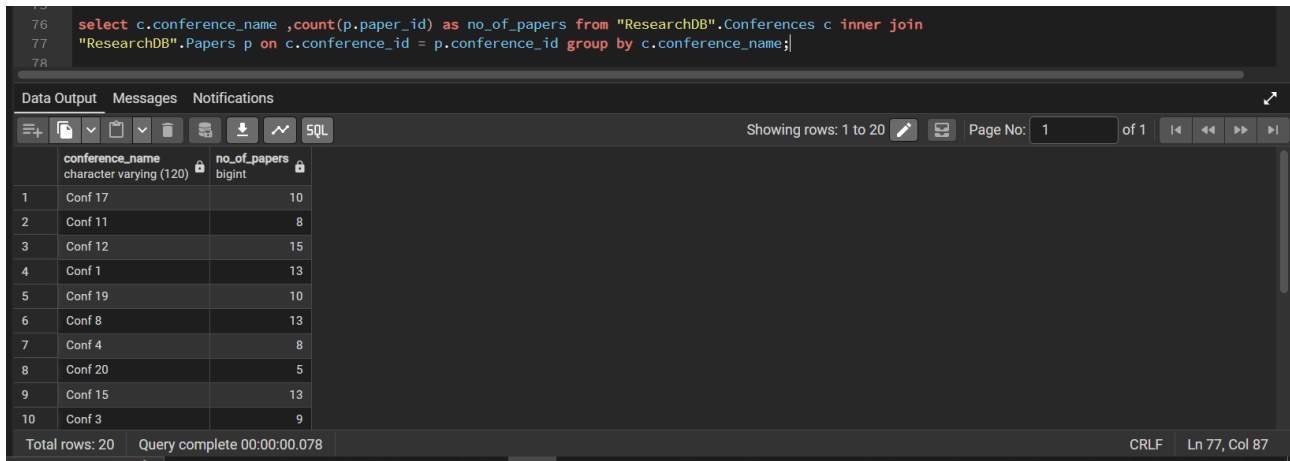
Total rows: 400 Query complete 00:00:00.144 CRLF Ln 71, Col 87

Tuple count = 400

Q5. Display all conferences and the number of papers published in each conference.

```
select c.conference_name , p.citation_count as no_of_pages from "ResearchDB".Conferences c  
inner join "ResearchDB".Papers p on c.conference_id = p.conference_id order by p.citation_count;
```

Tuple count = 20



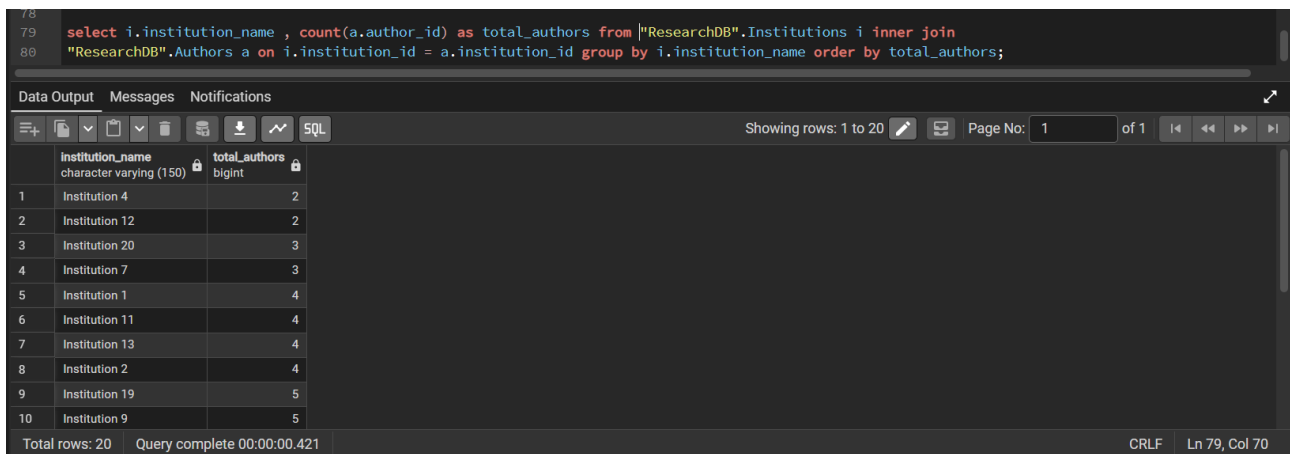
```
select c.conference_name , count(p.paper_id) as no_of_papers from "ResearchDB".Conferences c inner join  
"ResearchDB".Papers p on c.conference_id = p.conference_id group by c.conference_name;
```

	conference_name character varying (120)	no_of_papers bigint
1	Conf 17	10
2	Conf 11	8
3	Conf 12	15
4	Conf 1	13
5	Conf 19	10
6	Conf 8	13
7	Conf 4	8
8	Conf 20	5
9	Conf 15	13
10	Conf 3	9

Total rows: 20 Query complete 00:00:00.078 CRLF Ln 77, Col 87

Q6. Display all institutions and the total number of authors in each institution.

```
select i.institution_name , count(a.author_id) as total_authors from "ResearchDB".Institutions i  
inner join "ResearchDB".Authors a on i.institution_id = a.institution_id group by i.institution_name  
order by total_authors;
```



```
select i.institution_name , count(a.author_id) as total_authors from "ResearchDB".Institutions i inner join  
"ResearchDB".Authors a on i.institution_id = a.institution_id group by i.institution_name order by total_authors;
```

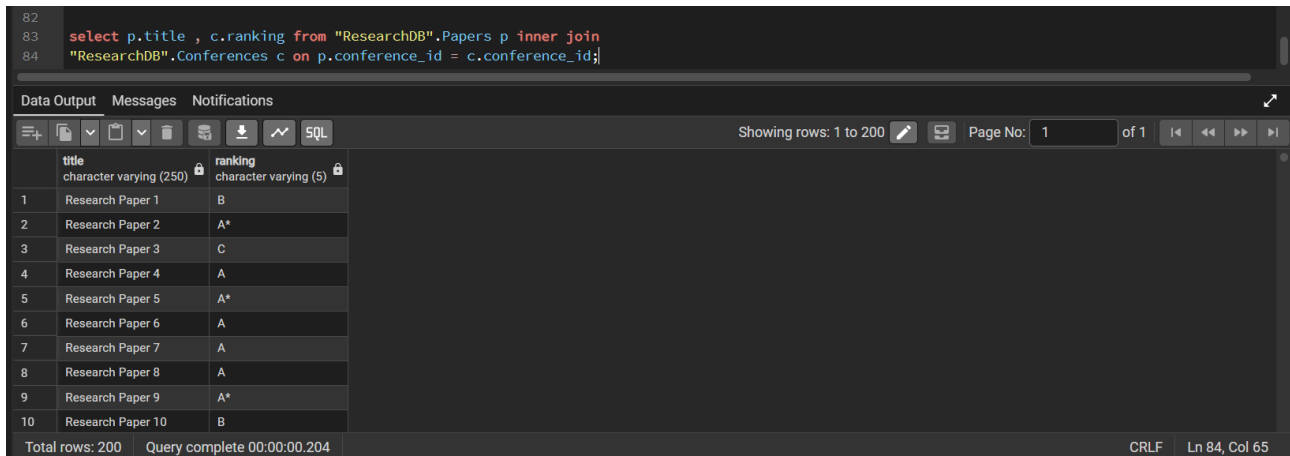
	institution_name character varying (150)	total_authors bigint
1	Institution 4	2
2	Institution 12	2
3	Institution 20	3
4	Institution 7	3
5	Institution 1	4
6	Institution 11	4
7	Institution 13	4
8	Institution 2	4
9	Institution 19	5
10	Institution 9	5

Total rows: 20 Query complete 00:00:00.421 CRLF Ln 79, Col 70

Tuple count = 20

Q7. Display the paper title and conference ranking.

```
select p.title , c.ranking from "ResearchDB".Papers p inner join "ResearchDB".Conferences c on
p.conference_id = c.conference_id;
```



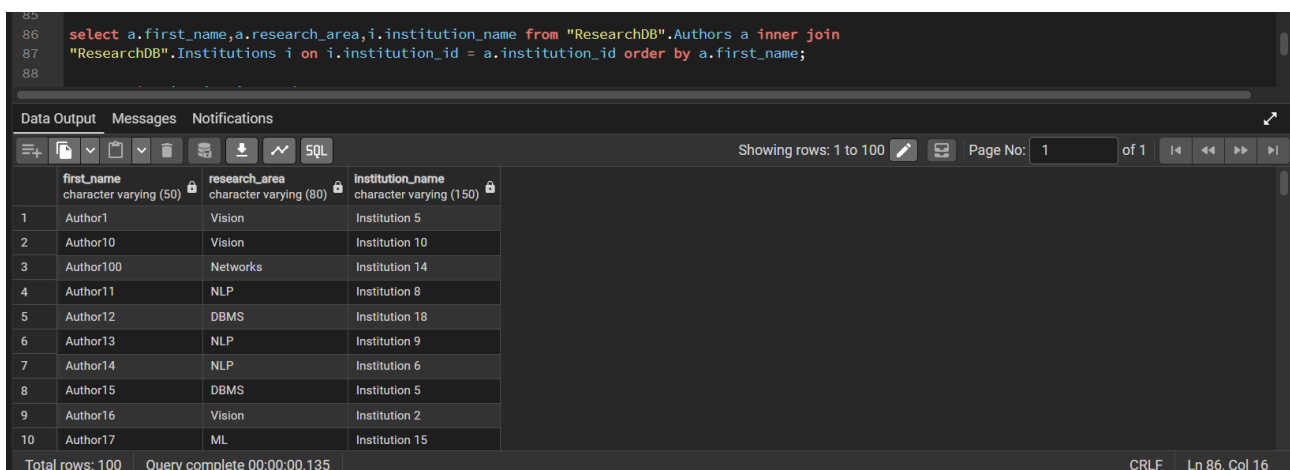
The screenshot shows a SQL query execution interface. The query is: `select p.title , c.ranking from "ResearchDB".Papers p inner join "ResearchDB".Conferences c on p.conference_id = c.conference_id;`. The results are displayed in a table with two columns: `title` (character varying (250)) and `ranking` (character varying (5)). The table shows 10 rows of data, numbered 1 to 10. The status bar at the bottom indicates "Total rows: 200" and "Query complete 00:00:00.204".

	title	ranking
1	Research Paper 1	B
2	Research Paper 2	A*
3	Research Paper 3	C
4	Research Paper 4	A
5	Research Paper 5	A*
6	Research Paper 6	A
7	Research Paper 7	A
8	Research Paper 8	A
9	Research Paper 9	A*
10	Research Paper 10	B

Tuple count = 200

Q8. Display the names of authors along with their research areas and institution names.

```
select a.first_name,a.research_area,i.institution_name from "ResearchDB".Authors a inner
join "ResearchDB".Institutions i on i.institution_id = a.institution_id order by a.first_name;
```



The screenshot shows a SQL query execution interface. The query is: `select a.first_name,a.research_area,i.institution_name from "ResearchDB".Authors a inner join "ResearchDB".Institutions i on i.institution_id = a.institution_id order by a.first_name;`. The results are displayed in a table with three columns: `first_name` (character varying (50)), `research_area` (character varying (80)), and `institution_name` (character varying (150)). The table shows 10 rows of data, numbered 1 to 10. The status bar at the bottom indicates "Total rows: 100" and "Query complete 00:00:00.135".

	first_name	research_area	institution_name
1	Author1	Vision	Institution 5
2	Author10	Vision	Institution 10
3	Author100	Networks	Institution 14
4	Author11	NLP	Institution 8
5	Author12	DBMS	Institution 18
6	Author13	NLP	Institution 9
7	Author14	NLP	Institution 6
8	Author15	DBMS	Institution 5
9	Author16	Vision	Institution 2
10	Author17	ML	Institution 15

Tuple count = 100

Q9. Create a view to display author names and their institutions.

```
create view institution_author as select a.first_name , i.institution_name from
"ResearchDB".Authors a inner join "ResearchDB".Institutions i on a.institution_id =
a.institution_id;
```

```
88
89 create view institution_author as
90 select a.first_name , i.institution_name from "ResearchDB".Authors a join
91 "ResearchDB".Institutions i on a.institution_id = i.institution_id;
92
93
```

Data Output Messages Notifications

CREATE VIEW

Query returned successfully in 118 msec.

Q10. Display all records from the view created in Q9.

```
select * from institution_author;
```

```
93
94 select * from institution_author;
95
96
```

Data Output Messages Notifications

Showing rows: 1 to 100 Page No: 1 of 1

	first_name character varying (50)	institution_name character varying (150)
1	Author1	Institution 5
2	Author2	Institution 13
3	Author3	Institution 18
4	Author4	Institution 12
5	Author5	Institution 15
6	Author6	Institution 16
7	Author7	Institution 18
8	Author8	Institution 13
9	Author9	Institution 17
10	Author10	Institution 10

Total rows: 100 Query complete 00:00:00.124 CRLF Ln 96, Col 1

Tuple count = 100

Q11. Create a view to display paper title, publication year, and citation count.

create view q11 as select p.title , p.publication_year , p.citation_count from "ResearchDB".Papers p;

```
97 create view q11 as
98 select p.title , p.publication_year , p.citation_count from "ResearchDB".Papers p;
99
```

Data Output Messages Notifications

CREATE VIEW

Query returned successfully in 90 msec.

Q12. Drop the view created in Q11.

drop view q11;

```
101
102 drop view q11;
```

Data Output Messages Notifications

DROP VIEW

Query returned successfully in 84 msec.

Q13. Insert a new conference using a transaction.

Q14. Update the citation count of a paper and commit the transaction.

```
BEGIN; UPDATE "ResearchDB".papers SET citation_count = citation_count + 50 WHERE  
paper_id = 5; COMMIT;
```

```
104  
105 BEGIN;  
106 UPDATE "ResearchDB".papers  
107 SET citation_count = citation_count + 10  
108 WHERE paper_id = 1;  
109 COMMIT;
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 92 msec.

Q15. Delete a keyword using a transaction and roll back the changes.

```
BEGIN;  
delete from "ResearchDB".Keywords  
where keyword_name = 'Keyword1';  
rollback;
```

```
111  
112 BEGIN;  
113 delete from "ResearchDB".Keywords  
114 where keyword_name = 'Keyword1';  
115 rollback;
```

Data Output Messages Notifications

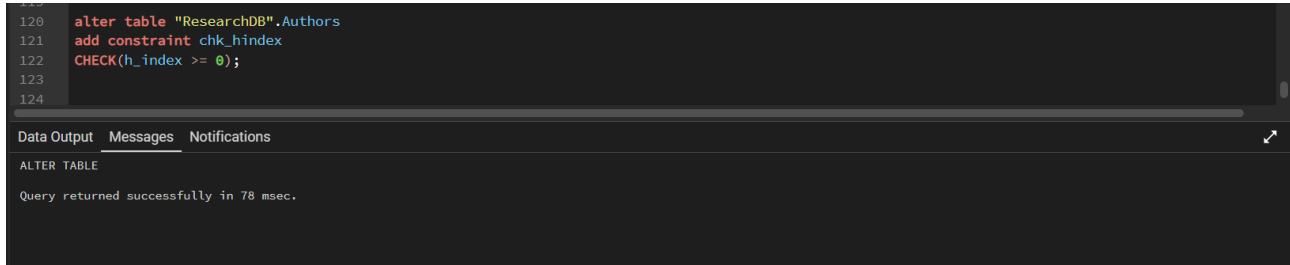
Showing rows: 1 to 40 Page No: 1 of 1

	keyword_id [PK] integer	keyword_name character varying (60)
1	1	Keyword1
2	2	Keyword2
3	3	Keyword3
4	4	Keyword4
5	5	Keyword5
6	6	Keyword6
7	7	Keyword7
8	8	Keyword8
9	9	Keyword9
10	10	Keyword10

Total rows: 40 Query complete 00:00:00.074 CRLF Ln 116, Col 1

Q16. Add a CHECK constraint to ensure the h-index is non-negative.

```
alter table "ResearchDB".Authors  
add constraint chk_hindex  
CHECK(h_index >= 0);
```



The screenshot shows a SQL IDE interface. The top pane displays the following SQL code:

```
120 alter table "ResearchDB".Authors  
121 add constraint chk_hindex  
122 CHECK(h_index >= 0);  
123  
124
```

The bottom pane has three tabs: "Data Output", "Messages", and "Notifications". The "Messages" tab is selected and shows the following output:

```
ALTER TABLE  
  
Query returned successfully in 78 msec.
```